



University of San Carlos - Talamban Campus

Department of Computer Information Sciences and Mathematics

In Partial Fulfillment of the Requirements of
IT 5101N - Business Analytics

IT 5101N Final Project - Phase 1

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Phase 1: Business Problem & Dataset Selection

Business Domain and Context

This project focuses on **e-commerce sales performance and customer purchasing behavior**. E-commerce businesses generate large amounts of transaction data from customer orders, product purchases, shipping activity, and regional sales. By analyzing this data, the business can better understand which products sell the most, which categories generate the highest revenue, how sales change over time, and which customer segments contribute to overall performance.

The purpose of this project is to use business analytics to turn e-commerce transaction data into useful insights. The analysis will focus on sales trends, product and category performance, regional patterns, customer demographics, shipping outcomes, and basic predictive relationships. These insights can help support better sales strategy, inventory planning, customer targeting, and operational decision-making.

Dataset Description

Item	Description
Dataset Name	E-Commerce Sales and Customer Insights Dataset
Source	GitHub: https://github.com/jeffersonmwatati/E-commerce-Sales-Analysis-Hackathon/blob/64527b0963ecab250ac30d546b2417a0cdfe6f42/raw-dataset-e-commerce-sales-and-customer-insights-dataset.csv
Format	CSV
Final Working File	Cleaned MySQL-ready CSV
Number of Records	1,000
Number of Attributes	20
Time Period	January 1, 2023 to January 3, 2024
Unit of Analysis	Each row represents one e-commerce transaction/order
Business Domain	E-commerce / Online Retail

The dataset contains transaction-level e-commerce sales records. It includes customer information, product details, order quantity, pricing, shipping fee, shipping status, and order date. The cleaned version also includes additional fields for age treatment and date-based analysis, such as `age_cleaned`, `age_group`, `age_model`, `year`, `month`, and `year_month`.

Data Preparation Summary

The data was cleaned using OpenRefine and additional validation checks to ensure that the dataset was suitable for business analytics and MySQL import. Before using the dataset for analysis, the file was cleaned and prepared to make it more suitable for business analytics and MySQL import. Column names were standardized into a database-friendly format, order dates were converted into a consistent YYYY-MM-DD format, and additional date fields such as `year`, `month`, and `year_month` were created to support trend analysis.

The Age column was treated carefully because some records had missing customer age values. Instead of deleting rows, the original age column was preserved, while a new `age_cleaned` column was created. Missing ages were recovered using the same Customer ID when possible. Remaining unresolved ages were placed into an Unknown age group for demographic analysis, while `age_model` was created for predictive modeling using median imputation. This allows the project to remain transparent while still supporting EDA, visualizations, and modeling.

Data Dictionary

Field Name	Data Type	Description
customer_id	Text/Categorical	Unique customer identifier. Customers may appear more than once if they placed multiple orders.
gender	Categorical	Gender of the customer.
region	Categorical	Region associated with the customer/order.
age	Numeric	Original customer age field. Some values are missing.
age_cleaned	Numeric	Cleaned age field after recovering missing values using the same Customer ID where possible.
age_group	Categorical	Age category used for customer demographic analysis.
age_missing_flag	Numeric/Flag	Indicates whether the original age value was missing.
age_model	Numeric	Age field prepared for predictive analysis using imputation where needed.
age_treatment_method	Text/Categorical	Explains how the age value was handled during cleaning.
product_name	Categorical	Name of the product purchased.
category	Categorical	Product category, such as Electronics, Accessories, or Wearables.
unit_price	Numeric	Price per unit of the product.
quantity	Numeric	Number of units purchased.
total_price	Numeric	Total transaction value.
shipping_fee	Numeric	Shipping cost charged for the order.
shipping_status	Categorical	Shipping outcome, such as Delivered, Returned, In Transit, or Unknown.

order_date	Date	Date when the order was placed.
year	Numeric	Year extracted from the order date.
month	Numeric	Month extracted from the order date.
year_month	Date/Text	Monthly period used for trend analysis.

Business Questions

Q1 — Descriptive Statistics

What are the key statistical properties of our numeric sales variables (unit price, quantity, total price, shipping fee), and are there any notable anomalies or distributional patterns?

Before any serious analysis, the business needs a baseline understanding of its own numbers. How much does a typical order cost? Is pricing consistent or all over the place? Are there unusually large or small orders that could skew decisions? This question establishes the foundation everything else builds on.

Main Variables: `year_month`, `total_price`

Q2 — Trend Analysis

How has monthly revenue trended across 2023, and are there any seasonal peaks or dips that the business should plan around?

No business operates the same way every month. This question helps the business spot whether revenue is growing, shrinking, or seasonal — so they can plan inventory, staffing, and promotions ahead of time instead of reacting after the fact.

Main Variables: `year_month`, `total_price`

Q3 — Cross-Tabulation Analysis

How does total revenue vary across product categories and customer regions simultaneously — which category-region combinations are strongest and weakest?

Not every product sells equally well in every region. A category might dominate nationally but be weak in one region, which could mean a gap in marketing, logistics, or customer preference. This question finds those blind spots so the business can target efforts more precisely.

Main Variables: *region, category, total_price*

Q4 — Drill-Down Analysis

Starting from total revenue, how does performance break down by category, then by product, and finally by shipping status — where are the hidden losses?

Top-line revenue can look healthy while losses are hiding underneath. A product might generate a lot of sales but also a lot of returns, quietly eating into profit. This question peels back the layers to find where revenue is actually being lost or at risk.

Main Variables: *category, product_name, shipping_status, total_price*

Q5 — Correlation and Predictive Analysis

Which numeric variables are most correlated with total revenue, and can we build a simple linear model to predict order value?

Understanding what drives revenue lets the business make smarter decisions. Should they push higher-priced products or focus on selling more units? The model gives a concrete, data-backed answer — and lets them estimate expected revenue for any given order scenario.

Main Variables: *unit_price, quantity, total_price, shipping_fee (correlation matrix) → then unit_price + quantity as predictors of total_price*

Ad Hoc Q6 — Follow-Up from Q2

After spotting a revenue dip in a specific month, which product category drove that dip — and did it recover?

Spotting a dip in Q2 is useful, but knowing what caused it is what makes it actionable. If Electronics dropped, that's a supply or demand issue. If Accessories dropped, it might be a promotions gap. Without this follow-up, the business knows something went wrong but not what to fix.

Main Variables: *year_month, category, total_price*

Ad Hoc Q7 — Follow-Up from Q3

Given that some region-category pairs underperform, does the customer age group explain the low spend in those segments?

A weak category-region combination could mean the product is irrelevant there — or it could mean the wrong customers are being reached. If Wearables underperform in a region dominated by older customers, the product isn't the problem; the targeting is. This question helps the business decide whether the fix is the product, the pricing, or the marketing.

Main Variables: *age_group, category, total_price*

Scope of the Analysis

This project focuses on descriptive business analytics using cleaned e-commerce transaction data. The analysis will examine sales performance, product/category behavior, regional differences, customer demographics, shipping outcomes, and basic predictive relationships. The goal is to provide insights that can support sales strategy, inventory decisions, customer targeting, and operational improvements.